



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

DIGITAL TWINS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Emerging Technologies

TOTAL: 10 QUESTIONS

Q1 What is Digital Twins and what problem in Emerging Technologies does it solve? **Threat Model**

Q6 How does Digital Twins handle reliability and high availability? **Availability**

Q2 What are the core components or concepts in Digital Twins? **Twins**

Q7 What observability does Digital Twins provide out of the box? **Box**

Q3 How does Digital Twins compare to its main alternatives? **Alternatives**

Q8 What are common performance bottlenecks in Digital Twins? **Twins**

Q4 How do you get started with Digital Twins for a new project? **Project**

Q9 What security best practices apply when running Digital Twins? **Testing**

Q5 What configuration options most affect Digital Twins' behavior in production? **SSO / IdP**

Q10 What mistakes do teams commonly make when adopting Digital Twins? **Tuning**



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Digital Twins is a tool or platform

Mitigation

Digital Twins is a tool or platform that automates or simplifies a specific concern in the Emerging Technologies domain, reducing manual effort and operational complexity for engineering teams.

A6 Digital Twins provides HA through leader election,

Observability

Digital Twins provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Digital Twins has key building blocks that

Primitives

Digital Twins has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Digital Twins exposes metrics (Prometheus endpoint or

Automation

Digital Twins exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Digital Twins trades off simplicity against feature

Hardening

Digital Twins trades off simplicity against feature richness differently than its alternatives. Choose Digital Twins when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Digital Twins via its package manager

Gotchas

Install Digital Twins via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Digital Twins with the principle of

Assessment

Run Digital Twins with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.